

October 22, 2026

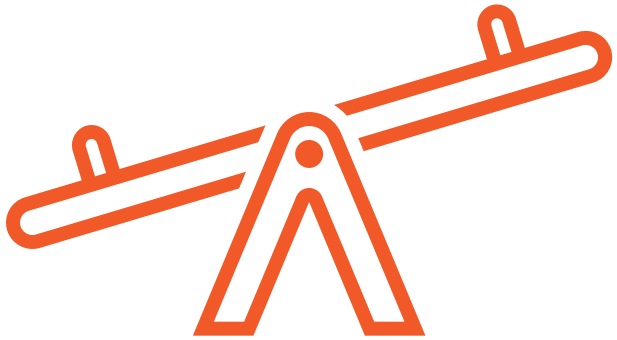
Session 2: Tariff Design, Cost Allocation, and Consumer Protection

RETA Dialogues

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Protecting Consumers

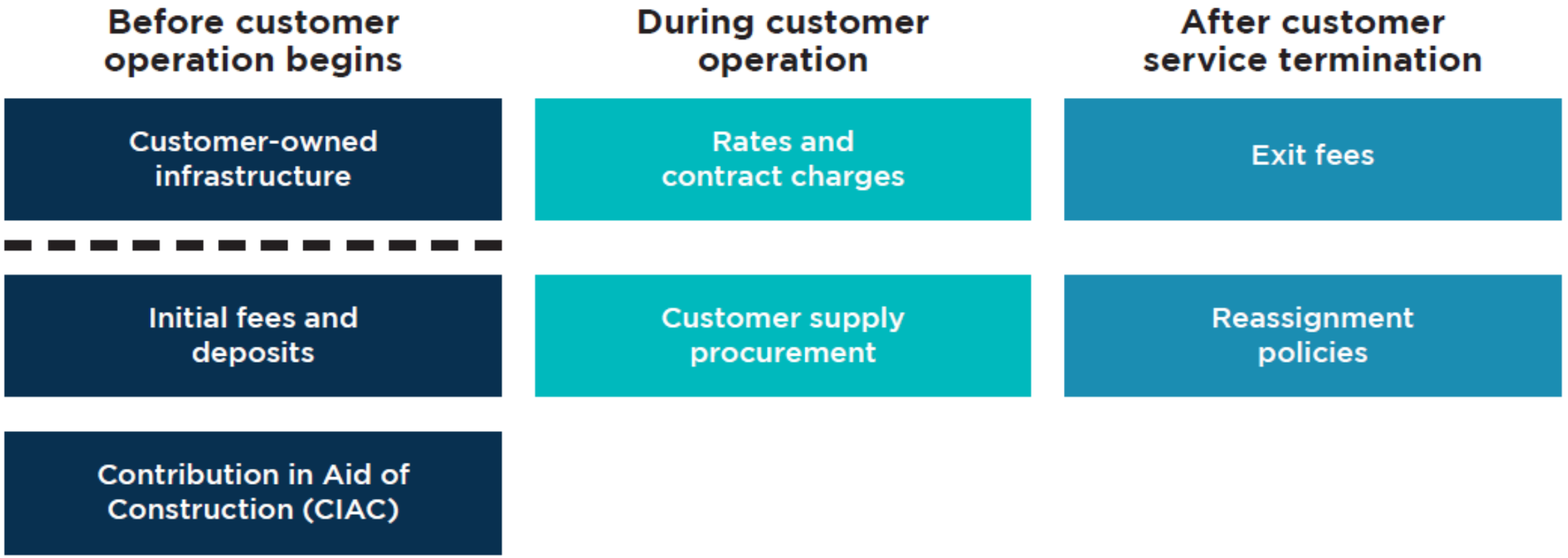
- Data centers are one form of load growth, albeit one with outsized impacts and unique characteristics
- Understanding the affordability impacts of data centers requires a careful look at the incremental costs and other impacts and a comparison with revenues and other benefits



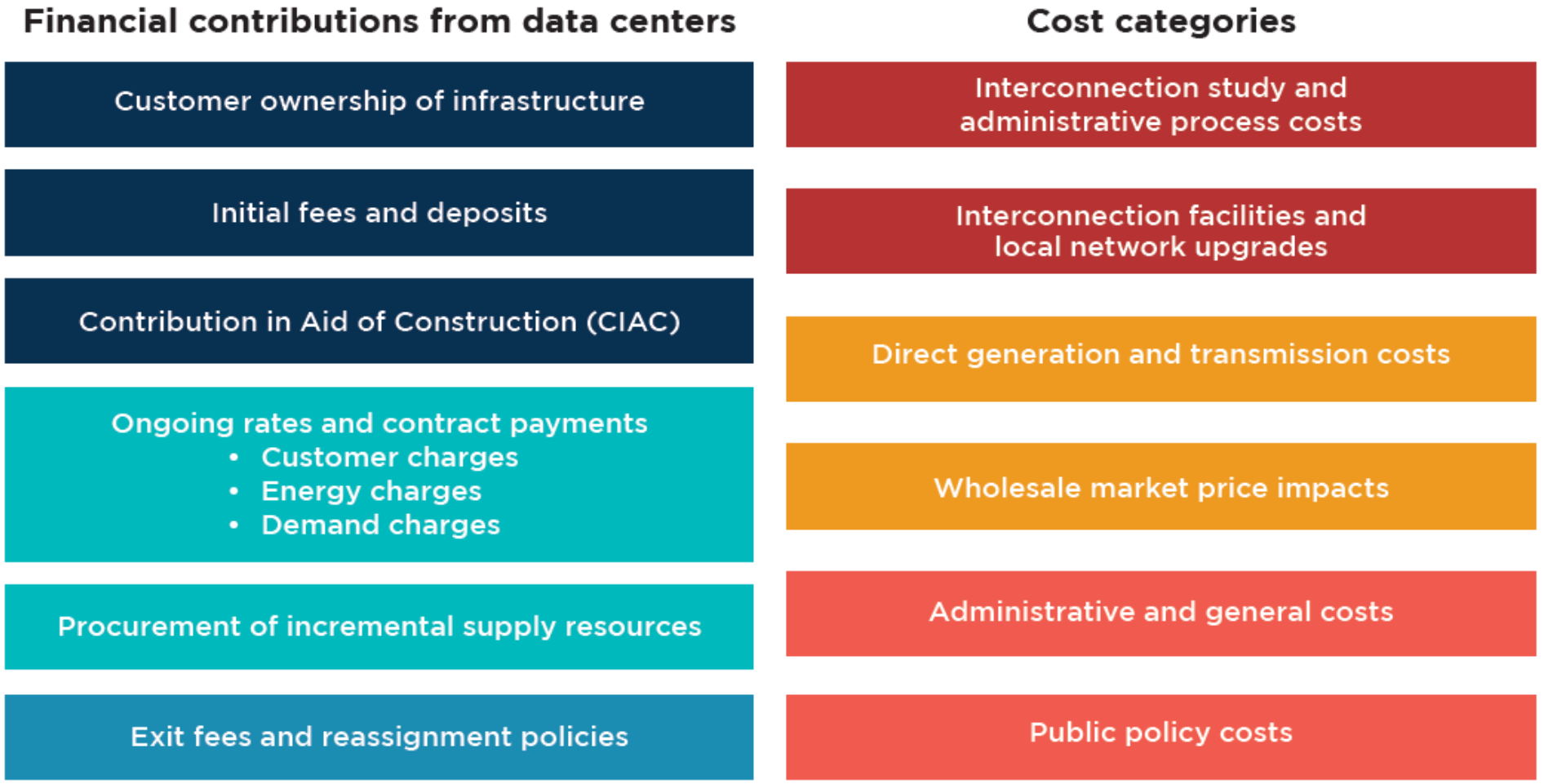
Key Factors Determine How Load Growth Impacts Rates

- Existing customers may pay more when marginal costs are higher than the additional revenue from the new customer/load
  - New load has high marginal costs
  - Rates for growing load are low
  - New load is speculative or could exit quickly
- Prior to new investments, previously unexpected load growth has a range of downsides
  - Expensive power plants generate more often
  - Higher line losses
  - Greater reliability risks

Full Lifecycle of Financial Contributions



Balancing contributions and costs



The Affordability Puzzle

Planning & Investments



Improving forecasts

- Scrutinizing data center demand decreases speculation
- Increases confidence that any new infrastructure will serve new load



Keeping costs low

- Considering low-cost and low-risk resources on equal footing
- Unlocking demand flexibility benefits

Regulatory Frameworks & Rates



Shaping the landscape

- Structural reforms make good results more likely
- Business model reform turns utility into partner



Allocating costs and risk

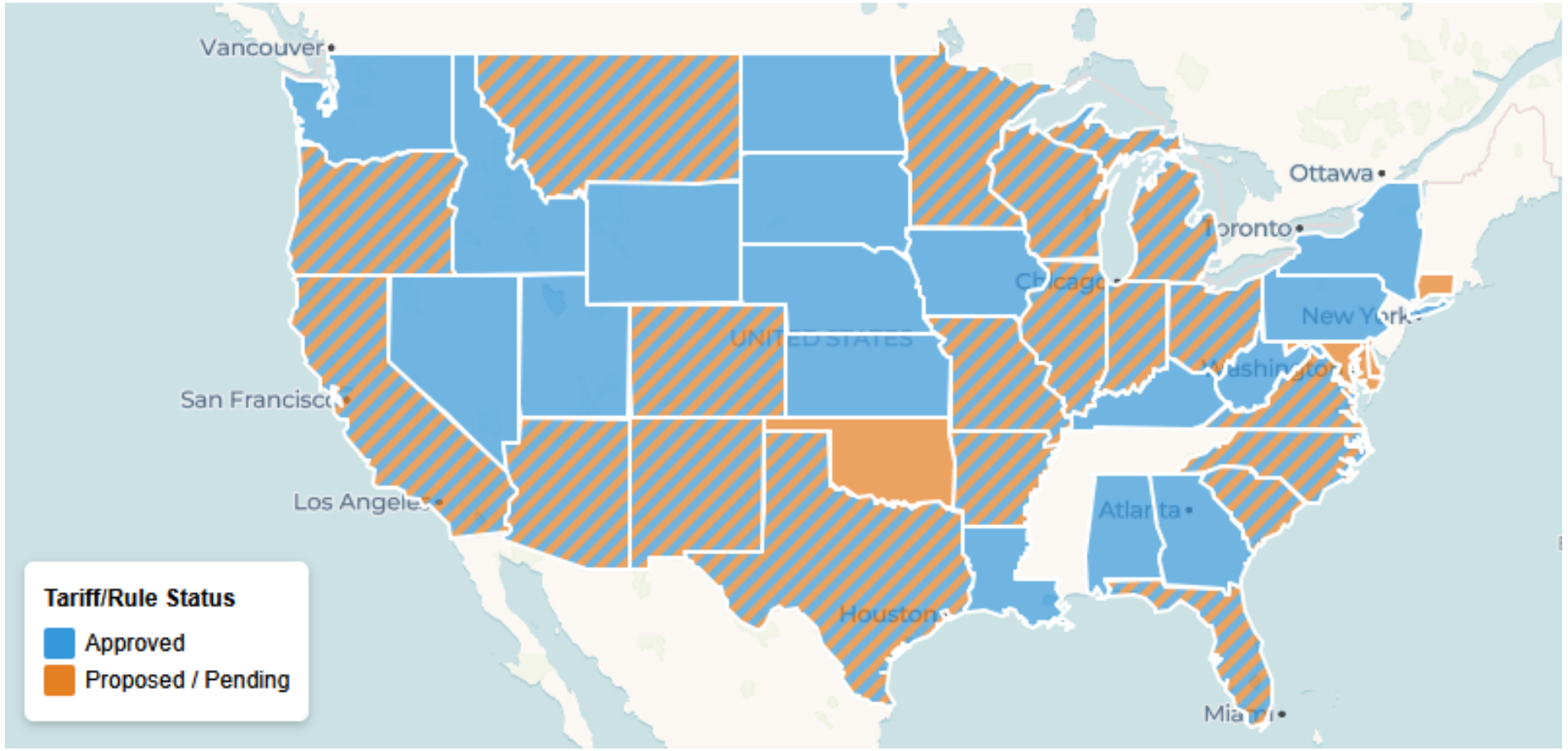
- Understanding the revenue life cycle
- Improving core cost allocation methods



Advanced rate design

- Balancing revenue certainty and efficient marginal cost pricing

Large load tariffs are well-established



Smart Electric Power Alliance and North Carolina Clean Energy Technology Center (2026). DELTa: Database of Emerging Large Load Tariffs. https://sepapower.org/large-load-tariffs-database/

Evolving tariff practices

Established practice
Stable or emerging practice

1. Define eligibility

- Minimum demand threshold
- Minimum load factor
- Aggregation rules

2. Ensure revenue

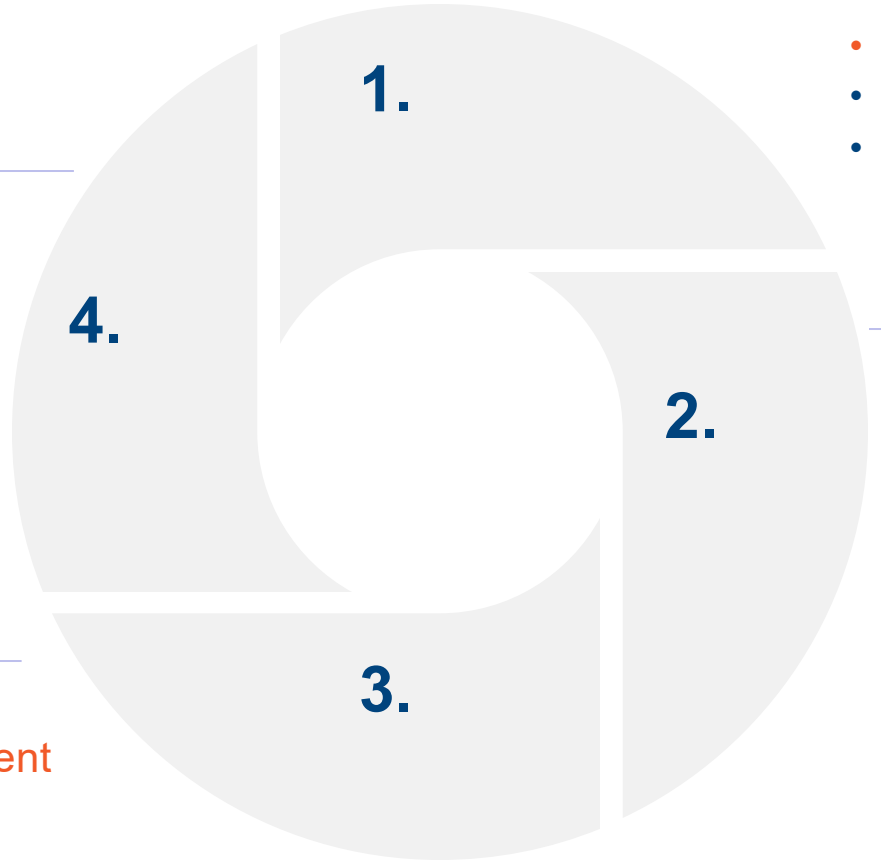
- Minimum contract duration
- Credit rating or collateral
- Take-or-pay requirements
- Load ramp period
- Price premium
- Hold harmless
- Resizing or reassigning capacity
- Exit fee

3. Allocate costs

- Direct assignment
- Customer-specific resource procurement

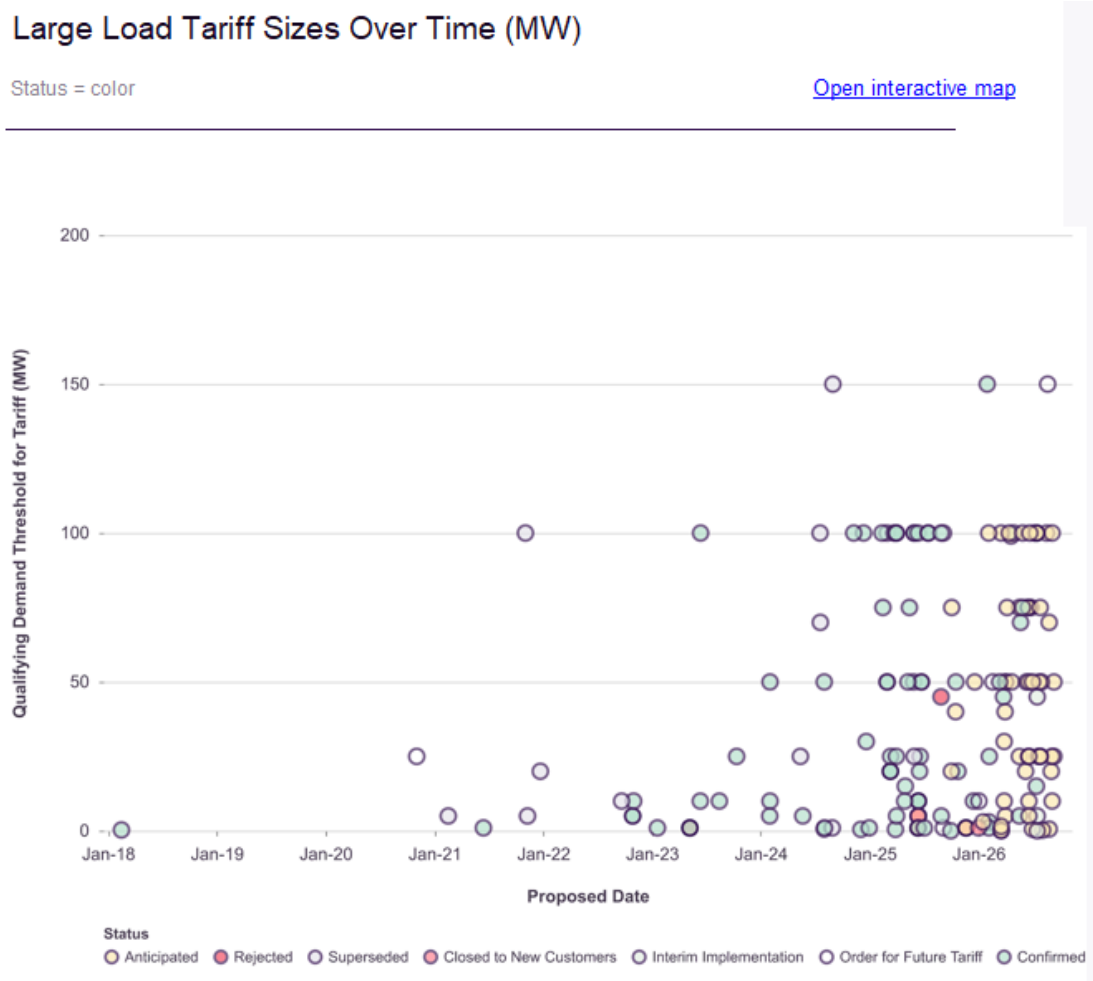
4. Improve planning

- Study requirement
- Load forecast requirement



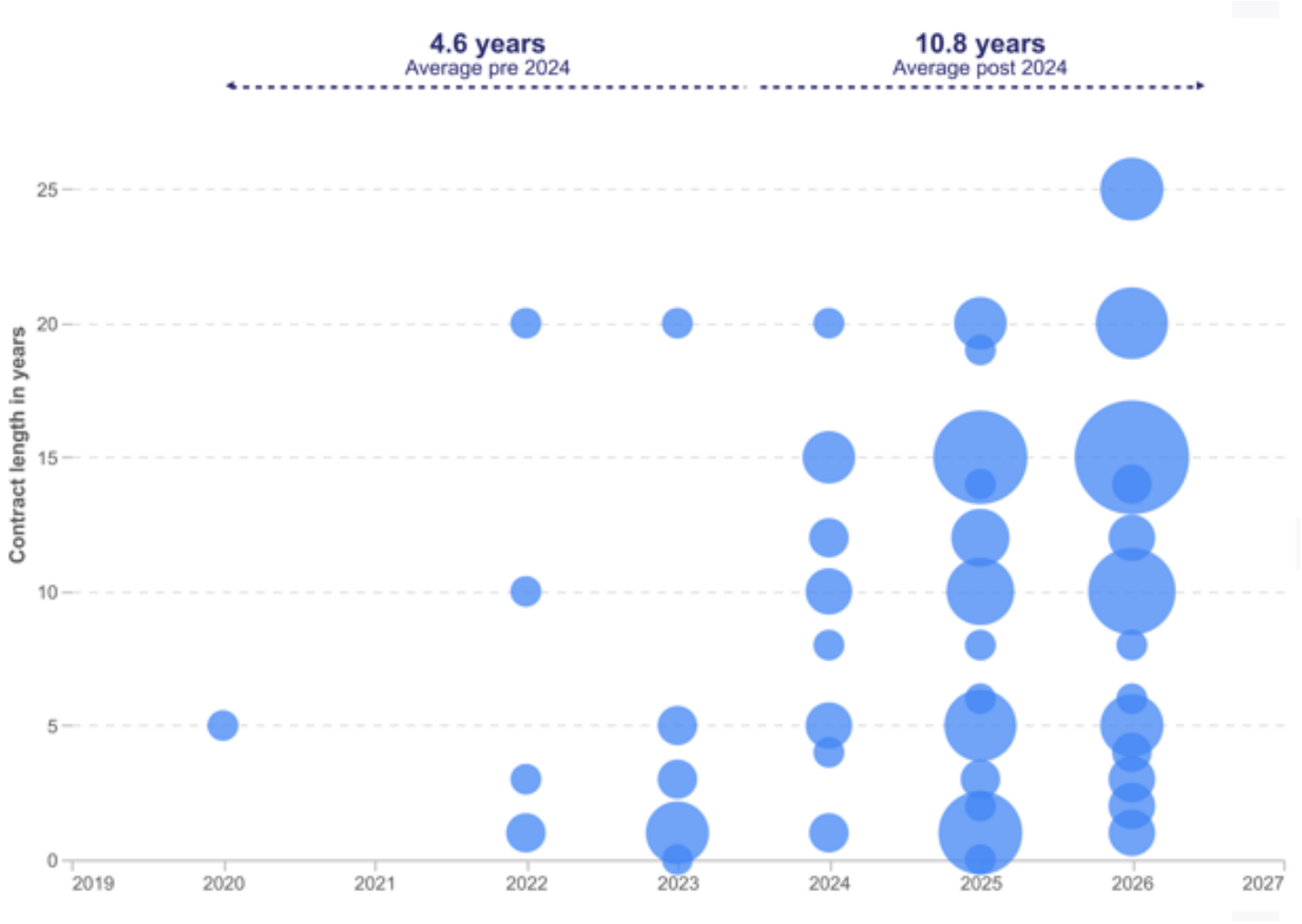
Source: LBNL, Electricity Rate Designs for Large Loads: Evolving Practices and Opportunities 2026 Update

Eligibility Threshold



Source: Halcyon, Large Load Tariff Tracker Sept 22, 2026 Update

Minimum Contract Duration



Source: Halcyon, Large Load Tariff Tracker Sept 22, 2026 Update

Take-or-pay

Demand charge

Recover capacity-related costs

Minimum billing demand

Sets floor on demand used for billing

Median: 80% of customer's contract demand

Minimum bill

Customer pays a minimum bill, regardless of actual consumption

Drawback: Disincentivizes customer flexibility and energy efficiency

Customer-specific procurement

- Behind-the-meter resources: allow or require BTM to serve some or all customer demand, reduce capacity, or provide flexibility
- Customer-led procurement: includes utility-managed power purchase agreements & Bring Your Own Capacity
- Load flexibility: allow or require customers to modify consumption during specified system conditions

Load flexibility is a key tool

- Reduce grid strain during peak demand → avoided generation capacity, transmission, and distribution infrastructure
- Increased utilization of existing grid infrastructure can spread legacy costs across more kWs and kWhs, reducing rates
- Can complement generation variability (intermittent resources, maintenance, outages)

Varying flexibility potential – data centers

Flexibility source	Flexibility lever	Underlying mechanism	Flexibility type	Constraints
IT system load	Data centre fleet workload scheduling	Computation tasks are shifted to another data centre where power system conditions are more favourable		Reserved to fleet operators (mostly hyperscale)
	Smart workload scheduling	Some computation tasks can be delayed and run during periods of lower stress on the grid		Only for time independent tasks (AI training, batch jobs)
Cooling system load	Temperature control management	Due to thermal inertia, cooling power can be modulated while keeping server temperatures within acceptable limits		Lower thermal inertia of newer server generations
On-site power systems	Back-up generators	During periods of high grid stress, backup generators can take over, so the data centre's power demand is no longer supplied by the grid (shedding)		Predominantly powered by fossil fuels
	Uninterruptible Power Systems (UPS)	The UPS system, designed to power the data centre while backup generators start during an outage, can also be used to provide ancillary services to the grid		Operator reluctance (fear of reduced reliability)
	Battery Energy Storage Systems (BESS) management	For reliability or under specific connection contracts, data centres can be co-located with BESS, which can be operated as a fully flexible asset: charging during periods of energy abundance and covering the data centre load during times of system stress		Requires BESS system design or retrofit

Spatial load shifting

second hour day

Temporal load shifting

Load shedding

Source: Agora Energiewende (2026) based on Deloitte: *Value and enablers of data centre flexibility in Europe*, 04/2026, https://www.agora-energiewende.org/publications/value-and-enablers-of-data-centre-flexibility-in-europe

About RAP

Regulatory Assistance Project (RAP)® is an independent, global NGO advancing policy innovation and thought leadership within the energy community.

Learn more about our work at raponline.org

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